
Q13 Continuous Manufacturing of Drug Substances and Drug Products Guidance for Industry

**U.S. Department of Health and Human Services
Food and Drug Administration
Center for Drug Evaluation and Research (CDER)
Center for Biologics Evaluation and Research (CBER)**

**March 2023
ICH**

Q13 Continuous Manufacturing of Drug Substances and Drug Products Guidance for Industry

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FOREWORD

The International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH) has the mission of achieving greater regulatory harmonization worldwide to ensure that safe, effective, and high-quality medicines are developed, registered, and maintained in the most resource-efficient manner. By harmonizing the regulatory expectations in regions around the world, ICH guidances have substantially reduced duplicative clinical studies, prevented unnecessary animal studies, standardized safety reporting and marketing application submissions, and contributed to many other improvements in the quality of global drug development and manufacturing and the products available to patients.

ICH is a consensus-driven process that involves technical experts from regulatory authorities and industry parties in detailed technical and science-based harmonization work that results in the development of ICH guidances. The commitment to consistent adoption of these consensus-based guidances by regulators around the globe is critical to realizing the benefits of safe, effective, and high-quality medicines for patients as well as for industry. As a Founding Regulatory Member of ICH, the Food and Drug Administration (FDA) plays a major role in the development of each of the ICH guidances, which FDA then adopts and issues as guidance to industry.

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Q13 Continuous Manufacturing of Drug Substances and Drug Products

Guidance for Industry¹

This guidance represents the current thinking of the Food and Drug Administration (FDA or Agency) on this topic. It does not establish any rights for any person and is not binding on FDA or the public. You can use an alternative approach if it satisfies the requirements of the applicable statutes and regulations. To discuss an alternative approach, contact the FDA office responsible for this guidance as listed on the title page.

PART I: CONTINUOUS MANUFACTURING OF DRUG SUBSTANCES AND DRUG PRODUCTS

I. INTRODUCTION (1)²

A. Objective (1.1)

This guidance describes scientific and regulatory considerations for the development, implementation, operation, and lifecycle management of continuous manufacturing (CM). Building on existing International Council for Harmonization (ICH) Quality guidances, this guidance provides clarification on CM concepts and describes scientific approaches and regulatory considerations specific to CM of drug substances and drug products.

B. Scope (1.2)

This guidance applies to CM of drug substances and drug products for chemical entities and therapeutic proteins. It is applicable to CM for new products (e.g., new drugs, generic drugs, biosimilars) and the conversion of batch manufacturing to CM for existing products. The principles described in this guidance may also apply to other biological/biotechnological entities.

CM involves the continuous feeding of input materials into, the transformation of in-process materials within, and the concomitant removal of output materials from a manufacturing process. While this description may apply to an individual unit operation (e.g., process chromatography, tableting, perfusion cell culture), this guidance focuses on the integrated aspects of a CM system in which two or more unit operations are directly connected. In this context, any changes made

¹ This guidance was developed within the Expert Working Group (Quality) of the International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH) and has been subject to consultation by the regulatory parties, in accordance with the ICH process. This document has been endorsed by the ICH Assembly at *Step 4* of the ICH process, November 2022. At *Step 4* of the process, the final draft is recommended for adoption to the regulatory bodies of the ICH regions.

² The numbers in parentheses reflect the organizational breakdown of the document endorsed by the ICH Assembly at *Step 4* of the ICH process, November 2022.

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in a unit operation of a CM system may have impact on downstream and upstream unit operations (e.g., back pressure resulting in forward mixing) and output material quality.

Fundamental aspects of CM that are generally not specific to technology, dosage form, or molecule type are described within the main body of this guidance. Annexes are provided to augment the main body of the guidance by providing illustrative examples and considerations specific to certain modalities (e.g., chemical entities, therapeutic proteins), technologies, and production methods (e.g., integration of drug substance and drug product manufacturing). The examples and approaches described in these annexes are illustrative, and alternative approaches can be used. Topics that are broadly applicable to both CM and batch manufacturing are not in the scope of this guidance, and other existing ICH guidances should be used as appropriate.

In general, FDA's guidance documents do not establish legally enforceable responsibilities. Instead, guidances describe the Agency's current thinking on a topic and should be viewed only as recommendations, unless specific regulatory or statutory requirements are cited. The use of the word *should* in Agency guidances means that something is suggested or recommended, but not required.

II. CM CONCEPTS (2)

A. Different Modes of CM (2.1)

CM can be applied to some or all unit operations in a manufacturing process. Examples of CM modes include:

- A manufacturing approach in which some unit operations operate in a batch mode while others are integrated and operate in a continuous mode
- A manufacturing approach in which all unit operations of a drug substance or drug product manufacturing process are integrated and operate in a continuous mode
- A manufacturing approach in which drug substance and drug product unit operations are integrated across the boundary between drug substance and drug product to form a single CM process (i.e., the drug substance is continuously formed and processed into the drug product through integrated unit operations)

A manufacturing approach may incorporate surge lines or tanks to maintain a constant flow of material inputs and outputs in any mode of CM described above.

B. Batch Definition (2.2)

The ICH guidance for industry *Q7 Good Manufacturing Practice Guidance for Active Pharmaceutical Ingredients* (September 2016)³ definition of a *batch* is applicable to all modes of

³ We update guidances periodically. For the most recent version of a guidance, check the FDA guidance web page at <https://www.fda.gov/regulatory-information/search-fda-guidance-documents>.

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CM, for both drug substances and drug products. Based on this definition, the size of a batch produced by CM can be defined in terms of one of the following:

- Quantity of output material
- Quantity of input material
- Run time at a defined mass flow rate

Other approaches to define batch size are possible, if scientifically justified based on the characteristics of the CM process and Good Manufacturing Practice.

A batch size can also be defined as a range. For example, a batch size range can be established by defining a minimum and maximum run time.

III. SCIENTIFIC APPROACHES (3)

A. Control Strategy (3.1)

The development of a control strategy for CM is enabled by a holistic approach, considering aspects specific to CM (discussed below), principles outlined in the ICH guidances for industry *Q7, Q8(R2) Pharmaceutical Development* (November 2009), *Q10 Pharmaceutical Quality System* (April 2009), *Q11 Development and Manufacture of Drug Substances* (November 2012) and quality risk management described in the ICH guidance for industry *Q9 Quality Risk Management* (June 2006).

1. State of Control (3.1.1)

A state of control (ICH Q10) is a condition that provides assurance of continued process performance and product quality. The condition may vary, depending on the mode of CM and the specific process steps. For example, a state of control can be demonstrated for CM processes when a set of parameters (e.g., process parameters, quality attributes) are changing within specified ranges, but the processes are not necessarily in a steady state. Elements of the control strategy monitor the state of control and, when necessary, take appropriate actions to maintain control of the process. It is important to have mechanisms in place to evaluate the consistency of unit operations and the system, and to identify when parameters drift or trend within the specified range. In addition, the root cause of drift or trend, such as variation of inputs, equipment fatigue, or aging of materials, should be identified. For example, in a therapeutic protein process, a change in the elution profile may be due to resin aging.

2. Process Dynamics (3.1.2)

Knowledge of process dynamics is important to maintain a state of control in CM. Specifically, understanding the impact of transient events helps to identify risks to material quality and to develop an appropriate control strategy (see section III.A.5 (3.1.5) for process monitoring and control considerations). Transient events that occur during CM operation may be planned (e.g., process start-up, shutdown and pause) or unplanned (e.g., disturbances) and can be described through time, process parameter and quality attribute values.

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Process dynamics should be characterized to understand how output material quality is impacted by transient events. This characterization could be done by determining properties such as residence time distribution (RTD). RTD characterizes the time available for material transport and transformation, and is specific to the process, composition/formulation, material properties, equipment design and configuration, etc. Additionally, understanding RTD and process dynamics enables the tracking of material and supports the development of sampling and diversion strategies, where applicable. Furthermore, such understanding is of importance from a process performance perspective. For example, changes in process dynamics or RTD may impact process characteristics, such as conversion/yield and impurity formation in the manufacture of drug substances.

Process dynamics should be characterized over the planned operating ranges and anticipated input material variability using scientifically justified approaches. Appropriate methodologies (e.g., RTD studies, *in silico* modeling with experimental confirmation) should be used to understand the impact of process dynamics and its variation on material transport and transformation. These methodologies should not interfere with the process dynamics, and the characterization should be relevant to the commercial process. For example, when conducting RTD studies, the tracer used to replace a constituent of the solid or liquid stream should have highly similar flow properties as those of the constituent replaced, be inert to the other components of the process, and not alter how processed materials interact with equipment surfaces. Step testing by making changes to the quantitative composition of the process stream (e.g., increments of a constituent) is another useful technique to determine the RTD and avoid the addition of an external tracer to the process. Other approaches can be used; the approach taken should be justified.

3. Material Characterization and Control (3.1.3)

Material attributes can impact various aspects of CM operation and performance, such as material feeding, process dynamics, and output material quality. Understanding the impact of material attributes and their variability on process performance and product quality is important for the development of the control strategy. Input materials may warrant evaluation and control of attributes beyond those typically considered for a material specification used in batch manufacturing. For example:

- In a solid dosage form process, particle size, cohesiveness, adhesiveness, hygroscopicity, static charge, or specific surface area of drug substances and excipients may impact the feeding of powders and material flow through the system.
- In a chemically synthesized drug substance process, viscosity, concentration, or the multiphase nature of the feed may impact flow properties or conversion.
- In a therapeutic protein (e.g., monoclonal antibody) process, lot-to-lot variability of cell culture media or feed components may impact cell culture performance, process performance, or process consistency.

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4. Equipment Design and System Integration (3.1.4)

The design of equipment and their integration to form a CM system affect process dynamics, material transport and transformation, output material quality, etc. When developing a CM process and its control strategy, it is important to consider the characteristics of the integrated systems in addition to the individual equipment that can affect process performance. These include the system's ability to maintain an integrated flow, manage potential disruption to CM operations, and complete the intended transformation of the material stream within the respective planned operational ranges of the equipment. In addition, transfer steps should also be evaluated, including RTD, for the purposes of integration. Examples of design considerations are given below:

- Design and configuration of equipment (e.g., compatibility and integrity of equipment components for the maximum run time or cycles; geometry of constituent parts to promote the desired transformation; spatial arrangement of equipment to facilitate material flow, maintenance, and avoid build-up or fouling)
- Physical connections and digital control interfaces between equipment (e.g., use of a surge tank between two unit operations to mitigate temporary differences in mass flow rates)
- Locations of material diversion and sampling points (e.g., selection of locations for a diverter valve and sampling probe without disturbing material flow and transformation)

Furthermore, appropriate design or selection of equipment for a CM process may enable process simplification (e.g., through a reduction in the number of unit operations), facilitate process monitoring and material diversion, and improve process capability and performance. For example, in a drug substance process, reactor design can effectively reduce formation and build-up of impurities, resulting in fewer purification steps. Similarly, for therapeutic protein drug substance manufacturing, system design can enable process intensification and reduce cycle times.

5. Process Monitoring and Control (3.1.5)

Process monitoring and control support the maintenance of a state of control during production and allow real-time evaluation of system performance. Common approaches to process monitoring and control—including establishment of target setpoints and control limits, design space, and specifications for attributes being measured—are applicable to CM.

Process analytical technology (PAT) (ICH Q8(R2)) is well-suited for CM. Example applications include in-line ultraviolet (UV) flow cells to monitor therapeutic protein concentration, in-line near-infrared spectroscopy to assess blend uniformity or water content, and online HPLC to monitor conversion of a chemical reaction. The use of PAT enables disturbances to be detected in real time. Therefore, CM is readily amenable to automated process control strategies based on, for example, active process control such as feedforward or feedback process control.

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Principles of control strategy as described in the ICH guidances for industry Q8(R2) and Q11 are applicable to CM processes.

An appropriate sampling strategy is an important aspect of process monitoring and control. The variables monitored, monitoring method and frequency, amount of material sampled (either physical sampling or data sampling using in-line measurement), sampling location, statistical method, and acceptance criteria depend on the intended use of the data and process dynamics. The intended use of data may include detection of rapid changes such as disturbances, assessment of quality of a batch when real-time release testing (RTRT) (ICH Q8(R2)) is used, or analysis of process trends or drifts. Further important considerations are the avoidance of measurement interference with the process as well as the impact of physical sampling on the material stream potentially affecting state of control. Assessment of risks associated with data gaps (e.g., PAT recalibration, refill of a feeding system, failure of system components) should inform whether contingency methods are warranted.

6. Material Traceability and Diversion (3.1.6)

CM processes may include periods when nonconforming materials are produced, for example, during system start-up and shutdown or when disturbances are not appropriately managed and mitigated. The ability to detect and divert potential non-conforming material from the product stream during production is an important characteristic of CM and should be considered when developing the control strategy.

Understanding the RTD and process dynamics of individual unit operations and integrated systems over planned operating conditions enables tracking of the distribution of materials over time and allows input materials to be traced throughout production. Material traceability, understanding how disturbances affect material quality, and the use of appropriate measurements (e.g., PAT) allow for real-time determination of when to start and stop material collection or diversion. The amount of material diverted can be influenced by several factors, such as process dynamics, RTD, control strategy, severity (e.g., magnitude, duration, frequency) of the disturbances, and location of the sampling and diversion points. Additionally, it is important that the diversion strategy accounts for the impact on material flow and process dynamics when material is diverted. Criteria should be established to trigger the start and end of the diversion period and restart of product collection.

7. Process Models (3.1.7)

Process models can be used for development of a CM process or as part of a control strategy for commercial production, including the diversion strategy. Process models may also be used to predict quality attributes in real time, enabling timely process adjustments to maintain a state of control. During development, process models can support the establishment of a design space by explaining how inputs (e.g., process parameters, material attributes) and outputs (e.g., product quality attributes) are related. Process models can enhance process understanding and reduce the number of experimental studies.

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For general considerations regarding models (including implications of model impact to validation requirements), refer to the ICH guidance for industry *Q8, Q9, & Q10 Questions and Answers* (July 2012). For CM applications, additional considerations are discussed below.

- A process model is specific to system design and configuration and relevant material properties.
- Model development requires an understanding of the underlying model assumptions (e.g., plug flow versus mixed flow systems) and when these assumptions remain valid. Risk assessments, sound scientific rationales, and relevant data inform the selection of model inputs and model formulation. It is important to determine the relevant inputs that affect the model performance, based on appropriate approaches such as sensitivity analysis.
- Model performance depends on factors such as mathematical constructs and the quality of model inputs (e.g., noise, variability of data). When setting acceptance criteria for model performance, the model's intended use and the statistical approaches that account for uncertainty in the experimental measurement and model prediction should be considered.
- Model validation assesses the fitness of the model for its intended use based on predetermined acceptance criteria using statistically sound approaches. Model validation activities are primarily concerned with demonstrating the appropriateness of the underlying model assumptions and the degree to which sensitivity and uncertainty of the model and the reference methods are understood.
- During commercial manufacture, model maintenance and monitoring of model performance should occur on a routine and ongoing basis considering variability that could impact the model and/or when a process change (e.g., input material, process parameter change) is implemented. Effective and efficient lifecycle management of models is enabled by risk assessment of the impact of a model change (e.g., optimization of model performance, change of the model's intended use, change of underlying model assumptions), considering the scope of model development and model validation criteria. Depending on the extent of a change and its impact on model performance, a model may need to be redeveloped and validated.

B. Changes in Production Output (3.2)

Several approaches to changing production output are discussed below; other approaches are also possible. For already-approved products manufactured using a CM process, it is important to justify the selected approach, assess risk and understand its impact on the overall control strategy and process performance, and as needed, update the control strategy to ensure risk control. Some changes may require process modification and process validation.

- **Change in run time with no change to mass flow rates and equipment:** Issues not observed over shorter run times may become visible as run time increases. Additional risks and constraints should be considered and may include, for example, process drift, increased equipment temperatures, material buildup, exceeding the performance limit of

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components (e.g., validated in vitro cell age, resin cycle number), material degradation, membrane or sensor fouling, and microbial contamination. Decreasing production output (below the longest run time previously validated) generally would not imply additional risks, given the same equipment, process and control strategy are used.

- **Increase mass flow rates with no change to overall run time and equipment:** The risks associated with this approach may impact output material quality and are related to changes in process dynamics and system capability to handle increased mass flow rates. Therefore, this approach may require re-evaluation and modification of the control strategy, including process parameters and controls, material traceability, RTD, sampling and diversion strategies to reduce risk.
- **Increase output through duplication of equipment (i.e., scale-out):** Considerations for two commonly used scale-out approaches are provided below.
 - *Replication of production lines (like-for-like):* Replicating the integrated CM production line (i.e., same equipment and setup as the original CM system) can be used to increase production output. The replicate production lines follow the same control strategy.
 - *Parallel unit operations on the same production line:* When only some unit operations are replicated on the same line, risks are associated with maintaining control across parallel unit operations. Aspects to consider are maintenance of uniform flow distribution among the parallel operations, synchronization and reintegration of parallel flow streams, changes to process dynamics, increasing the rate of other unit operations to handle the increased capacity, and material traceability.
- **Scale-up by increasing equipment size/capacity:** Depending on the process and equipment design, increasing production by increasing equipment size may be possible. General principles of equipment scale-up apply, as in the case of batch manufacturing. Because elements such as RTD, process dynamics, and system integration may change, various aspects of the control strategy may be impacted. The risks of scale-up and applicability of the original control strategy should be assessed at the new scale and the control strategy should be modified where needed.

C. Continuous Process Verification (3.3)

In CM, frequent process monitoring and control can be achieved through use of process parameters, PAT tools such as in-line/online/at-line monitoring and control, soft sensors, and process models. These tools allow real-time data collection for parameters relevant to process dynamics and material quality, and hence ensure the state of control for every batch. Additionally, since CM can facilitate changes to production output without increasing equipment size, there is an opportunity to generate development knowledge at the same scale intended for commercial manufacturing. These tools, together with the system design and the control

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strategy, facilitate early execution of process validation activities and the adoption of continuous process verification (ICH Q8(R2)) as an alternative approach to process validation.

IV. REGULATORY CONSIDERATIONS (4)

The dossier for a CM process should be in accordance with the ICH guidance for industry *M4: The CTD—Quality* (August 2001) (ICH M4Q). Some CM-specific considerations are provided below.

A. Description of Manufacturing Process and Process Controls (4.1)

In line with ICH M4Q, a sequential narrative description of the manufacturing process and process controls should be included in sections 3.2.S.2.2 and 3.2.P.3.3 of the Common Technical Document (CTD) and supported by pharmaceutical development data provided in CTD sections 3.2.S.2.6 or 3.2.P.2. In the case of CM, the information provided in sections 3.2.S.2.2 and 3.2.P.3.3 should be supplemented by the following, when applicable:

- A summary of start-up, shutdown, pause and restart procedures, as applicable, to describe how the integrated CM process functions
- The strategy for material collection and diversion including the criteria that should be met for product collection during routine manufacturing
- Process parameters related to continuous flow (e.g., mass flow rate(s) or flow rate range)
- When appropriate (e.g., movement of solid materials), a description of how the material is transported from one piece of equipment to another
- A flow diagram with the following aspects identified, when applicable:
 - Points of material entry into the process and material exit from the process (including material diversion and collection points)
 - Sequence of unit operations, including any surge lines or tanks
 - Direction of material movement through each process step
 - Clear indication of the continuous and batch process steps
 - Critical steps and location points at which process controls, intermediate tests, or final product controls are conducted (e.g., PAT measurement, feedforward, or feedback process control)

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- A suitably detailed description of any aspects of equipment design, configuration, and system integration that were shown during development to be critical to process control or to impact product quality

B. Control Strategy (4.2)

The control strategy of a CM process is designed to ensure that output materials made over time are of the desired quality. The control strategy should consider the elements discussed in section III (3) of the main body of this guidance. The dossier should describe the relevant controls and operational aspects (e.g., material diversion) used during manufacturing. Some aspects of the control strategy are discussed below.

- **Input material attributes:** Impact of input material attributes and their variability (e.g., inter-batch, different suppliers) on continuous processing and product quality should be assessed and proposed material attribute acceptable ranges should be justified when establishing the material specification. For input materials for which pharmacopeial requirements exist, characterization and control may extend beyond those requirements.
- **Process monitoring and control:** An appropriate description should be provided in the dossier to show a robust approach to process monitoring and maintaining a state of control. How the control system uses process parameters, in-process material attribute measurements, etc., to make process- and quality-related decisions (e.g., to pause the process or divert material) should be described. Other important aspects should be defined, such as the sampling strategy (e.g., location, sample size, frequency, statistical approach and criteria, and their relevance to the intended use), summary of the models if used (e.g., type of model used, strategies for model evaluation and performance monitoring), and the use of data in making in-process control decisions (e.g., to trigger material diversion).

The data analysis method used should appropriately detect the disturbances or variability in the CM process. For example, when data averaging is used, averaging across appropriate time intervals should be considered rather than data averaging across the entire CM run time. The time intervals should consider the relevancy of PAT measurement frequency to the RTD and process dynamics (i.e., process response time). Therefore, statistical sampling plans and data analysis should be described and justified.

- **System operation:** Procedures should be established and maintained on site for managing system start-up, shutdown, and pauses and for handling disturbances (see Annex V). Relevant approaches for these operations (e.g., handling disturbances) should also be described at an adequate level of detail in the dossier. The disposition of material impacted by transient and pause events should be justified, considering potential risks to output material quality (e.g., the impact of a disturbance as it propagates downstream).
- **Material diversion and collection:** The material diversion and collection strategy should be summarized and justified in the dossier. The strategy described should include the criteria for triggering material diversion, the basis for determining the extent of diverted

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materials, the conditions for resuming material collection, etc. Factors such as sampling frequency; RTD; and amplitude, duration and propagation of disturbances should be considered in developing the diversion strategy. The extent of diverted material should appropriately incorporate justified safety margins, considering the uncertainty of RTD and other measurements. Procedures for managing material collection, diversion, and disposition (e.g., quarantine, offline testing, investigations) do not need to be included in the dossier but should be maintained within the pharmaceutical quality system (PQS) (ICH Q10).

- **RTTRT:** RTTRT as described in the ICH Q8(R2) can be applied to an output material quality attribute(s). RTTRT is not a regulatory requirement for CM implementation. When RTTRT is proposed, the associated reference test method should be described. Development of the data collection approach for RTTRT implementation should include a risk assessment of how any lapses in data collection (e.g., recalibrating a near infrared (NIR) probe) may affect decisions relating to product quality. The proposed control strategy should include alternative or additional quality controls to mitigate any risks posed by data lapses. If the results from RTTRT fail or are trending towards failure, appropriate investigations should be conducted. Refer to ICH Q8, Q9, & Q10 *Questions and Answers* for discussion of models used as surrogates for traditional release testing methods.
- **Equipment and system integration:** Aspects of equipment design and system integration that are shown to be critical to output material quality and its control should be described and justified in the context of the overall control strategy.

A summary of the control strategy should be provided in CTD section 3.2.S.2.6 or 3.2.P.2.3 with links or references to the CTD sections that contain the detailed information to enable the understanding and evaluation of the manufacturing process and how it is controlled.

C. Batch Description and Batch Size (4.3)

The approach to define batch size (see examples in section II.B (2.2) and the proposed commercial batch size or range should be described in the dossier.

If a range is proposed, it should be justified, and the approach for achieving the range should be described (see section II.B (2.2)). Changes in batch size within the approved batch size range can be managed within the PQS. Any post-approval change beyond the approved range should be supported by data (section III.B (3.2)) and appropriately managed (i.e., prior approval or notification).

A suitable quantitative metric with acceptance criteria should be defined within the PQS to establish batch-to-batch consistency and system robustness. For example, when a batch size is defined by the amount of collected material, the amount of diverted materials relative to that of collected materials for each batch should be considered.

The actual intended size of a given batch should be defined before manufacturing begins.

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D. Process Models (4.4)

The level of detail provided in the dossier regarding model development, validation, and maintenance over the lifecycle should be commensurate with the model type and impact category. The process model should be specific for the defined system (e.g., equipment, layout, connections). Information to support models used as part of commercial manufacturing should be available during site inspection and maintained at or be accessible to the manufacturing site. Refer to ICH Q8, Q9, & Q10 *Questions and Answers* for regulatory expectations on process models.

E. Drug Substance and Drug Product Stability (4.5)

Regulatory expectations for the stability data package generally do not differ between CM and batch manufacturing modes (refer to the ICH guidances for industry *Q1A(R2) Stability Testing of New Drug Substances and Products* (November 2003) and *Q5C Quality of Biotechnological Products: Stability Testing of Biotechnological/Biological Products* (July 1996)). Batches used to generate primary stability data should be manufactured using a manufacturing process and equipment representative of the commercial process. Primary stability batches should incorporate the variability described in the ICH stability guidances (e.g., different drug substance batches).

Additionally, for chemical drug substances or drug products:

- Stability batches could be obtained with a single start-up/shutdown sequence provided the aforementioned variability is incorporated into the batches (e.g., by introducing different batches of drug substances in a sequential manner).
- Stability batches may be produced from shorter manufacturing runs, provided that (1) the control strategy, mass flow rate(s), and equipment used to generate stability batches are representative of the commercial process; and (2) it is demonstrated that a state of control is established and maintained when the process operates over the longer commercial run times. The concept of pilot scale batch for stability studies (e.g., at a minimum, one-tenth of a full production scale), as defined in other guidances (e.g., ICH Q1A(R2)), may not be necessarily applicable in this scenario.
- If modes of increasing production output discussed in section III.B (3.2) other than increasing run time are being used (e.g., increasing equipment size), applicants should justify their approach for defining primary stability batches. Applicants are encouraged to discuss their scale-up and primary stability batches approach with regulatory authorities.

F. Conversion of a Batch Process to CM (4.6)

Changing the manufacturing mode from batch to continuous necessitates the development of an appropriate control strategy, considering factors identified in section III (3). The output materials from the batch and continuous processes should have comparable quality. A science

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and risk-based approach should be used for establishing product comparability and assessing the need for additional bioequivalence, non-clinical and/or clinical studies, and stability data. Additional details regarding how to establish product comparability for therapeutic proteins can be found in the ICH guidance for industry *Q5E Comparability of Biotechnological/Biological Products Subject to Changes in Their Manufacturing Process* (June 2005).

Manufacturers should seek regulatory approval before implementing a conversion of an approved batch process to a CM process. Manufacturers can seek advice from the regulatory authority to gain clarification on the regulatory expectations and acceptability of their strategy and data package for the proposed changes (e.g., when potential changes in formulation are necessary to enable conversion to CM and the impact of these changes on product registration).

G. Process Validation (4.7)

The requirements for process validation as established by region are similar for CM and batch manufacturing processes. Approaches such as traditional process validation that uses a fixed number of validation batches or continuous process verification can be used.

When continuous process verification is used, the CM system performance and output material quality is continuously monitored, such that the real-time data collected demonstrate the maintenance of a state of control and production of output material with the desired quality for the run time duration. The use of continuous process verification should be justified based on the product and process understanding, system design, and overall control strategy. This justification should be provided in the dossier (see Table 1 below).

When a continuous process verification approach is used to support initial product launch, applicants should justify the appropriateness of the validation activities to provide confidence in the commercial manufacturing process.

H. Pharmaceutical Quality System (4.8)

PQS expectations are the same for batch and CM processes and should follow pertinent ICH guidances. One important operational aspect of CM is that nonconforming materials can be diverted from the rest of the batch when material traceability, process monitoring, and material diversion strategies are well established. Procedures for material diversion, when required, should be established under the PQS (see section IV.B (4.2)). An overarching plan or decision tree that describes how disturbances are managed for various types of material diversion should be maintained under the PQS. Diverted materials resulting from planned events (e.g., system start-up and shutdown) generally do not require investigation when the events meet established process performance criteria. Examples of approaches for managing disturbances are provided in Annex V. As described therein, when unexpected and/or frequent disturbances occur, appropriate investigation, root cause analysis, and corrective action and preventive action (CAPA) should be instituted.

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I. Lifecycle Management (4.9)

The principles and approaches described in the ICH guidance for industry *Q12 Technical and Regulatory Considerations for Pharmaceutical Product Lifecycle Management* (May 2021) are applicable to the lifecycle management of CM.

J. Submission of CM-Specific Information in the CTD (4.10)

The dossier should include information as outlined in ICH M4Q. Additional elements relevant to CM should also be provided in the dossier when applicable. Examples of these elements are listed in Table 1. In the case of integrated drug substance and drug product CM processes, some information and data, such as an integrated flow diagram, may be presented in CTD section 3.2.P with a cross reference in 3.2.S (see Annex IV for additional details).

Table 1: Recommended CM-Related Information in the CTD

CTD Section	Information and Data
3.2.S.2.6 3.2.P.2.3	Manufacturing Process Development • Summary of the overall process development • Summary and justification of the control strategy, with links or references to the CTD sections that contain comprehensive information on the control strategy elements, for example: ○ Strategy for system start-up, shutdown, and pause ○ RTD and material traceability information ○ Disturbance management (e.g., material diversion and collection strategy) ○ Sampling strategy, including sampling frequency ○ Equipment design and system integration aspects that are shown to be critical to output material quality ○ Process controls such as feedforward and feedback process control based on a measured attribute, if used ○ Development and justification of models[#] where used in the control strategy (e.g., process model used for feedforward control, model associated with a PAT analytical method used for in-process testing and/or real-time release) ○ Justification for range of batch sizes, where claimed

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3.2.S.2.2 3.2.P.3.2	Batch Definition Batch size or range, and approach to achieving the intended batch size or range
3.2.S.2.2 3.2.P.3.3	Description of Manufacturing Process and Process Controls - Narrative description of the commercial manufacturing process and flow diagram as described in ICH M4Q, clearly indicating which portion of the manufacturing process is continuous - Examples of CM-specific aspects of the commercial manufacturing process to be described are: - Equipment design and system integration aspects when critical to the output material quality - CM-related process parameters, controls, and criteria (e.g., input rates/mass flow rates, relevant feeder operating ranges), and location points at which process controls or testing is conducted - Location of active process controls, if used - Criteria for product collection, including strategy for diversion
3.2.S.2.4 3.2.P.3.4	Controls of Critical Steps and Intermediates - Critical process parameters - In-process testing (e.g., sampling frequency, sample size, analytical method) - Relevant information, parameters, and criteria associated with ensuring correct application of process models used as part of the control strategy[#] including contingency plan when the model is not available - Relevant information on active process controls, when applicable (e.g., limits of acceptability for controls that ensure monitored critical process parameters and critical quality attributes stay within desired ranges)
3.2.S.4 3.2.P.5	Control of Drug Substance or Drug Product - When models are associated with the analytical procedures for release testing of the drug substance or the drug product (e.g., NIR model, dissolution model): - Summary and justification of the model and the sampling strategy - Contingency testing and monitoring plans instituted for when the model is not available (e.g., when gaps in PAT data occur or in case of PAT equipment failure) - Analytical model validation information

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3.2.R	Regional Information • Validation data for high impact process models, if used • Continuous process verification scheme, if applicable
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[#] The purpose of a model can vary (e.g., testing the quality of in-process material, drug substance intermediate, drug substance or drug product, real-time release testing, process control). Not all categories are covered in this table; information relating to the models should be submitted in the appropriate CTD sections identified in ICH M4Q for these categories. For example:

- Models used for drug substance release testing in 3.2.S.4;
- Models used for in-process testing in 3.2.S.2.4 or 3.2.P.3.4;
- Model used for both in-process testing and real-time release testing in relevant 3.2.S.4 or 3.2.P.5 sections and incorporated by reference into the applicable Control of Critical Steps and Intermediates section.
- Models used only for process development in 3.2.S.2.6 and 3.2.P.2.3.

GLOSSARY

Active Process Controls: A system consisting of hardware and software architecture, mechanisms, and algorithms that automatically adjust a process to maintain the process output within a desired range. Examples include feedforward and feedback process controls.

Disturbances: Unplanned changes to process inputs beyond normal operating range or conditions (e.g., process parameter, material property, equipment condition, or environment) that are introduced into a system.

Diversion: Procedure in which materials are isolated and separated from the product stream in the manufacturing process.

Material Traceability: The ability to track materials throughout the manufacturing process.

Model Maintenance: A set of planned activities over the product lifecycle to monitor and sustain the model's performance to continually ensure its suitability for the intended and approved purpose.

Multivariate Statistical Process Control: The application of multivariate statistical techniques to analyze complex process data with potentially correlated variables (Ph. Eur.).

Process Dynamics: The response of a manufacturing process to changing inputs or conditions or transient events.

Residence Time Distribution (RTD): A measure of the range of residence times experienced by material passing through a specific process environment/vessel/unit operation.

Run Time: The time interval used to produce a quantity of output material.

Soft Sensors: A model that is used in lieu of physical measurement to estimate a variable or attribute (e.g., a quality attribute of material) based on measured data (e.g., process data). The model development, including selection of such data variables, is driven by comprehensive product and process understanding.

Steady State: A stable condition that does not change over time.

System: A manufacturing architecture that, in the context of CM, consists of individual pieces of equipment, their connections to one another, monitoring and control systems, and spatial layout.

Transient Events: A temporary condition in which a process goes through a dynamic change. This change may be due to a disturbance or an intentional alteration in the selected operating conditions (e.g., start-up, shutdown, changes from one operating condition to another).

Unit Operation: A basic step in a process. Unit operations involve a physical, chemical, or biological transformation such as: reaction, crystallization, filtration, blending, granulation, tableting, cell culture, purification, or virus inactivation.

REFERENCES

Guidances for Industry

ICH guidance for industry *M4Q: The Common Technical Document for The Registration of Pharmaceuticals for Human Use: Quality* (August 2001).

ICH guidance for industry *Q1A(R2): Stability Testing of New Drug Substances and Products* (November 2003).

ICH guidance for industry *Q5C: Quality of Biotechnological Products: Stability Testing of Biotechnological/ Biological Products* (July 1996).

ICH guidance for industry *Q5E: Comparability of Biotechnological/Biological Products Subject to Changes in Their Manufacturing Process* (June 2005).

ICH guidance for industry *Q6A: Specifications: Test Procedures and Acceptance Criteria for New Drug Substances and New Drug Products: Chemical Substances* (December 2000).

ICH guidance for industry *Q7: Good Manufacturing Practice Guide for Active Pharmaceutical Ingredients* (September 2016).

ICH guidance for industry *Q8(R2): Pharmaceutical Development* (November 2009).

ICH guidance for industry *Q8, Q9, & Q10 Questions and Answers* (July 2012).

ICH guidance for industry *Q9: Quality Risk Management* (June 2006).

ICH guidance for industry *Q10: Pharmaceutical Quality System* (April 2009).

ICH guidance for industry *Q11: Development and Manufacture of Drug Substances (Chemical Entities and Biotechnological/Biological Entities)* (November 2012).

ICH guidance for industry *Q12: Technical and Regulatory Considerations for Pharmaceutical Product Lifecycle Management* (May 2021).

Literature

Ph. Eur.: European Pharmacopoeia, available at <https://pheur.edqm.eu/home>.

PART II: ANNEXES

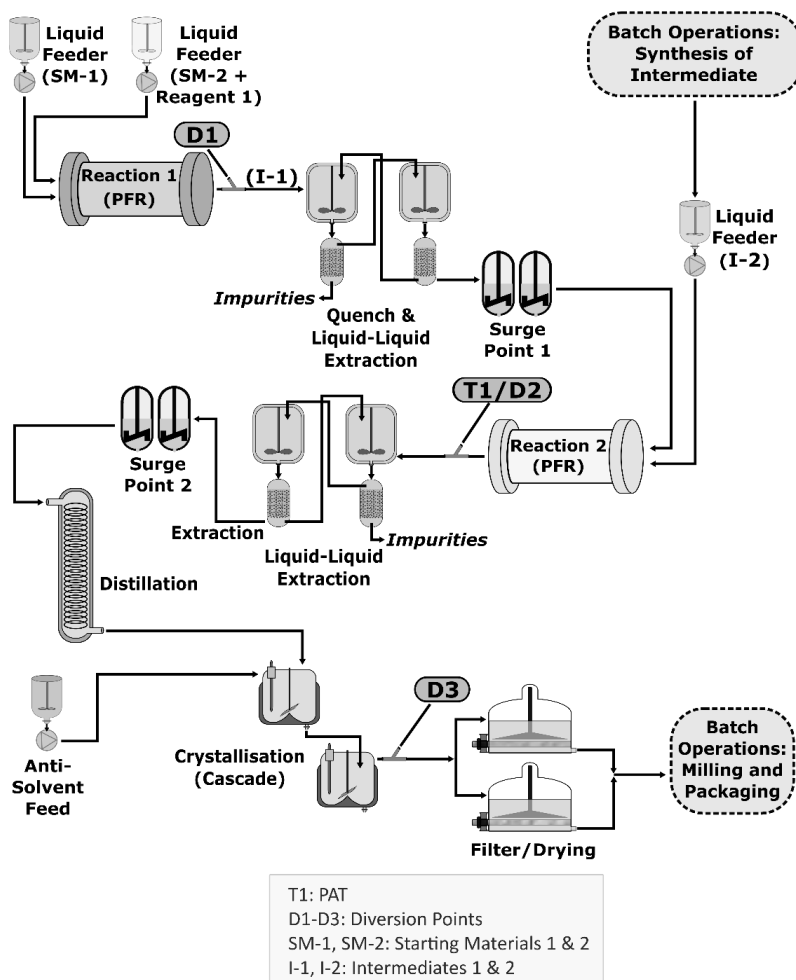
ANNEX I: CONTINUOUS MANUFACTURING OF DRUG SUBSTANCES (CHEMICAL ENTITIES)

I. INTRODUCTION AND EXAMPLE SYSTEM OVERVIEW (1)

This annex exemplifies one approach to implement CM of drug substances for chemical entities based on the scientific principles described in the main body of this guidance. The examples and approaches in this annex are illustrative, and alternative approaches can be used.

Figure 1 illustrates a drug substance manufacturing process containing both continuous and batch operations. It is not intended to represent a regulatory flow diagram. The continuous process segment consists of unit operations that can be characterized as having two plug-flow reactors (PFRs), two liquid phase extraction operations, continuous crystallization, and two agitated filter dryers. Manufacture of Intermediate 2 is performed in batch mode, as is final processing including milling and packaging. This annex focuses on the continuous elements of this process.

Figure 1: Example of a Drug Substance CM System for Chemical Entities



II. CONTROL STRATEGY AND OTHER TECHNICAL CONSIDERATIONS (2)

The CM system and its control strategy were designed to control parameters that impact the manufacture and quality of the drug substance, including impurity profile and physicochemical properties. The overall control strategy was developed in accordance with the main body of this guidance and the ICH guidances for industry Q7, Q8(R2), Q9, Q10, and Q11.

A. Equipment Design and Integration (2.1)

Within the continuous process segment in Figure 1 (section I Annex 1), the following processes occur:

- Reaction 1: Starting materials 1 and 2 are coupled in a PFR to produce Intermediate 1. Diversion Point D1 is located after the PFR to permit material diversion when PFR conditions are outside predefined acceptance criteria. The reaction is quenched as an integrated operation after the PFR, and unwanted by-products are removed by liquid-liquid extraction. The resultant solution (Intermediate 1) is used as an input for the second reaction without isolation.
- Reaction 2: Intermediate 1 and Intermediate 2 (prepared upstream through separate batch unit operations) are coupled in a second PFR to form the crude drug substance. The online PAT near the reactor exit (T1) monitors conversion of Intermediate 1 to the crude drug substance. Diversion Point D2 located after PAT is used to divert nonconforming material.
- Drug Substance Isolation: The crude drug substance is purified by liquid-liquid extraction and continuous two-stage crystallization. Distillation prior to crystallization provides desired concentration of the crude drug substance solution. Diversion Point D3 allows for material diversion at the crystallizer. The crystal slurry is filtered using two agitated filter dryers operating in an alternating fashion to enable continuous operation. The isolated drug substance is then milled using batch operations to achieve the desired particle size distribution prior to packaging.

Two surge points (each containing multiple surge tanks) are used: one before Reaction 2 and another before continuous crystallization. These are important components of the system design and control strategy, as they improve process robustness and mitigate temporary differences in mass flow rates by decoupling upstream and downstream operations.

The design of the overall system and each unit operation, along with the control strategy, optimize material quality. For example, PFR design elements (i.e., dimension and configuration) allow control of temperature, heat and mass transfer rates, and reaction time. These parameters were shown during development to be important to the drug substance impurity profile.

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B. Process Control and Monitoring (2.2)

Holistic controls used across Reactions 1 and 2 ensure consistent operations and quality of the resulting crude drug substance. The stoichiometry of Reaction 1 is controlled via control of concentrations and flow rates of the feeds. Conversion of starting materials to Intermediate 1 with minimal impurity formation is ensured through control of the reaction temperature and flowrate (i.e., reaction time). Reaction 2 is controlled through feedback control of the addition rate of Intermediate 2 based on the PAT measurement of residual Intermediate 1 levels. This ensures correct stoichiometry for that reaction and minimizes the impact of variability of the Intermediate 1 feed solution on drug substance purity. The PAT also measures levels of crude drug substance and impurities, which confirm successful operation of all preceding steps and consistent product quality. Liquid/Liquid extraction ensures appropriate purity control after Reaction 2.

RTD was used to develop a suitable strategy for disturbance detection, corrective actions, and material diversion. RTD characterization was based on mathematical modeling of all unit operations and surge points across the entire CM process over planned mass flow rates. The RTD was then confirmed through experimental tracer studies for appropriate segments of the commercial equipment. Decisions for triggering material diversion are based on comparing process parameters and PAT measurements to predefined acceptance criteria with timing and duration of diversion informed by the RTD. Importantly, the RTD is also used for material traceability purposes.

Understanding of process dynamics and its impact on quality attributes of material produced throughout the entire process was also used to guide start-up and shutdown strategies. For example, during start-up of Reactions 1 and 2, a small amount of Intermediate 1 or crude drug substance is diverted at Diversion Points 1 or 2, respectively, to allow those materials to reach the target concentrations before processing into subsequent operations. The criteria for diversion were established based on time considering the RTD. This approach was supported by development studies and appropriately confirmed in commercial process equipment. PAT monitoring after Reaction 2 provides additional verification that appropriate criteria have been met during start-up. Collection of material proceeds to the end of the process as subsequently described.

Sampling and process measurement needs were evaluated, considering relevant factors such as residence times (i.e., surge points, process dynamics, and the type and purpose of the measurement). The measurement frequency of the PAT at Reaction 2 enables detection of certain disturbances, informs process adjustments, and ensures timely diversion of material based on predefined criteria. The criteria for material diversion are based on the magnitude and duration of the disturbance, an understanding of RTD and process dynamics of downstream unit operations and surge points, and the impurity purging capability of the crystallization operation. As a result of this control strategy, all crude drug substance solution that enters continuous crystallization meets acceptable quality criteria and can be processed through the crystallizer.

Appropriate controls and monitoring requirements for the continuous crystallization were extensively investigated during development in similar, but smaller scale equipment and

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appropriately verified using commercial equipment. Process development included spiking studies using impurity-enriched feed solutions and intentional perturbations in process parameters (i.e., feed flow rates, their ratios, and temperatures). An evaluation of the encrusted solids in the crystallizer over extended run times demonstrated the solids were the same form and purity as the free-flowing drug substance slurry. The set of process parameters and ranges identified by these studies were appropriately scaled up. Implementation of these controls along with post-crystallization material tests (e.g., crystal form, purity) ensure consistent quality of the resulting drug substance throughout continuous crystallization and subsequent filtration.

The resulting material is milled using a batch operation to provide a drug substance of the appropriate particle size for use in drug product manufacturing. Procedures were developed to allow diversion of material at Diversion Point D3 in the event desired process conditions or material attributes are not met. Diversion of the drug substance from the crystallizer was found to be unnecessary either during start-up or shutdown.

C. Consideration of Other Controls (2.3)

Process robustness and performance over time are important considerations. A risk assessment was performed to ensure that adequate controls are in place to support the proposed run time (which can be up to several months). It identified a number of considerations and corresponding controls/measures. Examples are summarized in Table 2.

Table 2: Examples of Other Controls for Consideration

Consideration	Controls/Measures
Cleaning and fouling potential	• Establishment of a risk-based cleaning strategy, including understanding of the impact of build-up on drug substance quality • Additional monitoring to assess fouling and cleanliness (e.g., pressure sensors at the discharge of feed pumps, periodic visual checks for the continuous crystallizer) • Reduction of other risk factors (e.g., filtering feed streams to further reduce fouling risk)
Stability of in-process materials	• Hold times at key points in the process (e.g., feed streams; accumulated material at the surge points, reactors, and crystallizer) managed through batch record and process automation • Risk assessment of microbiological growth (i.e., negligible risk based on the nature of the process materials and conditions)
Calibration and potential for changes/drift in instrumentation	• Periodic checks at selected points (e.g., process parameter measurements for the PFR, system suitability for the PAT analyzer) • Dual sensors at selected locations (e.g., temperature probes for the PFR) so that appropriate corrective actions can be taken
Equipment maintenance	• Maintenance requirements for target run time • Use of redundant equipment (e.g., backup pumps) at key locations

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Additionally, specifications for input materials were evaluated during process development. There were no differences between batch and continuous processing for this example.

Collectively, the process understanding developed along with implementation of the various controls described provide a robust and reliable control strategy. This ensures consistent quality of the resulting drug substance including the impurity profile, physicochemical properties, and ability of the system to identify and appropriately react to unexpected events.

D. Process Validation (2.4)

The combination of process controls, online PAT measurements, comprehensive monitoring of process parameters and material attributes, and end-product testing results in a data-rich environment for this process. Together with system understanding generated during development, this combination of control strategy elements enabled the use of a traditional process validation for commercial product launch and continuous process verification to validate process changes over the product lifecycle.

A range of batch sizes was initially established based on material demands and the quantities of material necessary to match input needs of the final batch unit operations. The process was validated using a fixed number of batches. A single planned start-up and shutdown of the commercial CM system was used to manufacture the process validation batches. This approach was supported by the totality of evidence demonstrating the start-up and shutdown capabilities of the system. This evidence included development work on smaller equipment (with appropriate technical justification for being representative), commercial equipment and system qualification data, results of a pre-validation demonstration run, and extensive process monitoring of the CM system that can verify success of each start-up and shutdown in real time.

Subsequently, a continuous process verification approach was adopted after product approval to validate increases in batch size with extension of run time. The batch size increase was possible without impact to the equipment scale for downstream batch operations. This approach used a risk assessment for the longer run time, which concluded that process performance and material quality would not be impacted. Under the continuous process verification approach, data generated during the manufacture of each batch was used to support successful validation of that batch with the extended run time. This included information such as system performance monitoring and data logs, along with other controls that ensure material quality with appropriate detection and corrective action. Additionally, appropriate regulatory actions were taken to communicate this batch size increase with run time change and use of the continuous process verification approach.

III. REGULATORY CONSIDERATIONS (3)

Refer to section IV (4). In consideration of the specific CM process design, additional elements may need to be included in a dossier. For instance, in this example, the influence of surge points on the material diversion and collection strategy, including the fate of materials, was described.

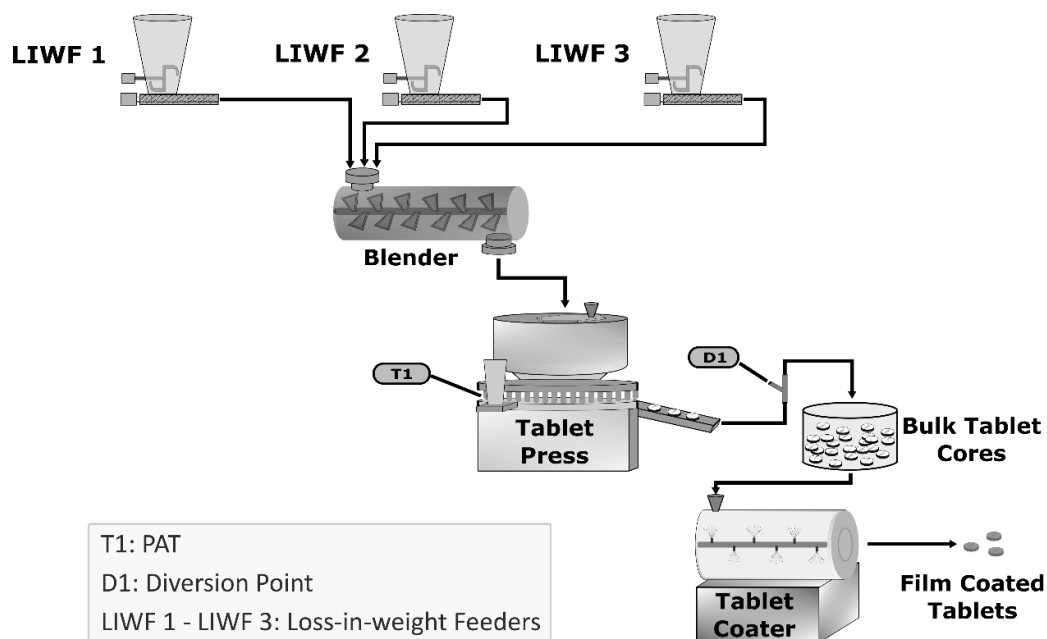
ANNEX II: CONTINUOUS MANUFACTURING OF DRUG PRODUCTS (CHEMICAL ENTITIES)

I. INTRODUCTION AND EXAMPLE SYSTEM OVERVIEW (1)

This annex exemplifies one approach to implementing CM for a tablet drug product based on the scientific principles described in the main body of this guidance. The examples and approaches in this annex are illustrative, and alternative approaches can be used. Specific considerations relating to the implementation of a continuous direct compression process for a chemical entity are presented.

Figure 2 illustrates a continuous direct compression process that consists of continuous feeding, blending, and tablet compression unit operations, with batch mode film coating. It is not intended to represent a regulatory flow diagram.

Figure 2: Example of a Tablet Drug Product CM System



A PAT tool using an NIR method monitors blend uniformity and triggers tablet diversion. Run time at a predefined mass flow rate is used to define the batch size range; in this case, the overall marketing demand requires batch sizes between 360 and 1080 kilograms of the drug product.

II. CONTROL STRATEGY AND OTHER TECHNICAL CONSIDERATIONS (2)

The CM system and its control strategy were designed to mitigate the impact of disturbances to ensure output material quality. The overall control strategy was developed in accordance with the main body of this guidance and the ICH guidances for industry Q8(R2), Q9, and Q10.

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A. Material Characterization and Control (2.1)

During process design and development, a quality-by-design approach was adopted that identified equipment and process parameters critical to control of the process. Furthermore, the relationships between material quality attributes and their impact on unit operations (particularly the loss-in-weight feeders (LIWFs) and blender) and product critical quality attributes (CQAs) were evaluated. Bulk density of the primary excipient and particle size distribution (PSD) of the drug substance were identified as critical to blend and content uniformity. A defined bulk density range and three-tier (d10, d50, d90) PSD specification were implemented for the excipient and drug substance, respectively.

B. Equipment Design and Integration (2.2)

Unit operations and system components (e.g., NIR probe) were designed or selected to mitigate the impact of disturbances on final product quality. The overall design principle is, where possible, to use gravity to move material. During system integration, the material flow was coordinated across all unit operations to avoid material accumulation or emptying. System mass balance was obtained through understanding of material flow (i.e., RTD) at the intended operating conditions of each unit operation. The impact of equipment design and operation on process dynamics was characterized by the RTD of individual unit operations, as well as the RTD of process segments between individual unit operations and the diversion point. The RTDs were determined by replacing the drug substance in the formulation with a tracer that has highly similar flow properties to those of the drug substance.

The following aspects of equipment design and integration were emphasized:

- **LIWF:** Feeder mass flow rates and their variability were characterized. LIWFs are controlled to deliver the theoretical amount of each input material per the formulation; it was demonstrated that the risks of minor variations to product composition were mitigated by blender mixing capability. Feeder mass flow rates were evaluated using design of experiment (DOE) studies and the proven acceptable ranges of target flow rates were defined. Modeling and statistical approaches were used to help determine the limits for the magnitude and duration of disturbances in mass flow rates, for which material diversion, operator investigation, or process stop are needed. These limits were visualized (e.g., funnel plots) to aid understanding. LIWFs operate in gravimetric mode unless they are refilling (volumetric mode). Refill aspects (e.g., duration and mass of refill) were evaluated to minimize the impact on feeding.
- **Blender:** A horizontal blender was selected for the CM system and the blender design was evaluated (e.g., paddle versus ribbon, number, and orientation of paddles in the blender, rotation speed). It was determined that a paddle blender is critical to ensure desired blend uniformity. Rotation speed, number and orientation of the paddles were evaluated for their impact on blend uniformity over the ranges studied, and the corresponding design space for the blending process was defined. RTD characterization provided information on the degree of forward and back mixing and disturbance

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propagation, and the RTD was used to define the material traceability and diversion strategy.

- **NIR probe**: The NIR probe was placed in the tablet press feed frame. The chosen NIR equipment met the PAT application requirements (e.g., analysis speed, sampling method, mass flow rate). Probe location and height were fixed; the impact of material build-up was evaluated and found not significant. The system intended for commercial production was used to generate data for the development, calibration, and validation of the NIR method.
- **Diversion point**: The RTD between the NIR probe and the diversion point was characterized using a tracer. The material diversion strategy links LIWF and NIR limits to the RTD between the LIWF and NIR as well as the RTD between the probe and diversion point, respectively.
- **Coater**: The mass in the coater corresponds to 1 hour of production. Coating was designed to be complete in 45 minutes; while coating occurs, the next aliquot of tablet cores is filled into the tablet hopper.

C. Process Controls and Monitoring (2.3)

In this system, the LIWFs may introduce fast dynamic disturbances. These disturbances may also occur during changes in operating conditions (e.g., during start-up or process pauses). Therefore, monitoring and control of these events are important elements of the control strategy. The control strategy includes NIR measurements, in-process controls (e.g., individual and total flow rates), process parameters including critical process parameters (e.g., blender rotation speed), and active process controls (e.g., feedback control of tablet weight). The sampling strategy for monitoring and control reflects the observed process dynamics, therefore ensuring adequate detectability of all relevant disturbances. Together, these aspects enable proactive control of the system and ensure continuous operation in a state of control and accurate material diversion to waste based on the predefined criteria. Unique codes are assigned to predefined batch segments to ensure material traceability and identification of conforming and non-conforming materials. Start-up/restart, pause/stop, and shutdown strategies for this example are defined in Table 3.

Table 3: Start-Up/Restart, Pause/Stop, and Shutdown Strategies

Action	Activity
Start-up/Restart	Material tracking and data collection begins; manufactured material is diverted until it meets the predefined acceptance criteria for material collection.
Pause/Stop	A process pause or stop is executed either manually or automatically, according to predefined criteria.
Shutdown	Material collection continues until manufactured material fails the predefined acceptance criteria, and then manufacturing ends.

D. Process Validation (2.4)

In this example, a continuous process verification approach was adopted, considering elements such as prior facility experience in implementing a similar CM process and control system (i.e., platform approach), availability of product-specific data arising from late-stage product development using the commercial equipment, the scale independence of the commercial process (i.e., batch size varies by run time), a comprehensive control strategy with high-frequency data collection, and the use of real-time data from every manufacturing run to further support continuous process verification. The control strategy provides real-time monitoring, trending, and prediction analysis through the use of NIR measurements, LIWF data, and other data sources arising from monitoring process parameters (e.g., blender torque), thus providing a high degree of assurance of real-time CM system stability and performance and output material quality. The continuous process verification approach, coupled with appropriate regulatory action for reporting manufacturing changes, was used to validate run time extensions beyond current experience.

III. REGULATORY CONSIDERATIONS (3)

Refer to section IV (4). In consideration of the specific CM process design, additional elements may need to be included in a dossier. For instance, in this example, elements that can significantly impact process dynamics and homogeneity (e.g., design space, number of paddles and their orientation in the horizontal paddle blender) were described.

ANNEX III: CONTINUOUS MANUFACTURING OF DRUG SUBSTANCES (THERAPEUTIC PROTEINS)

I. INTRODUCTION AND EXAMPLE SYSTEM OVERVIEW (1)

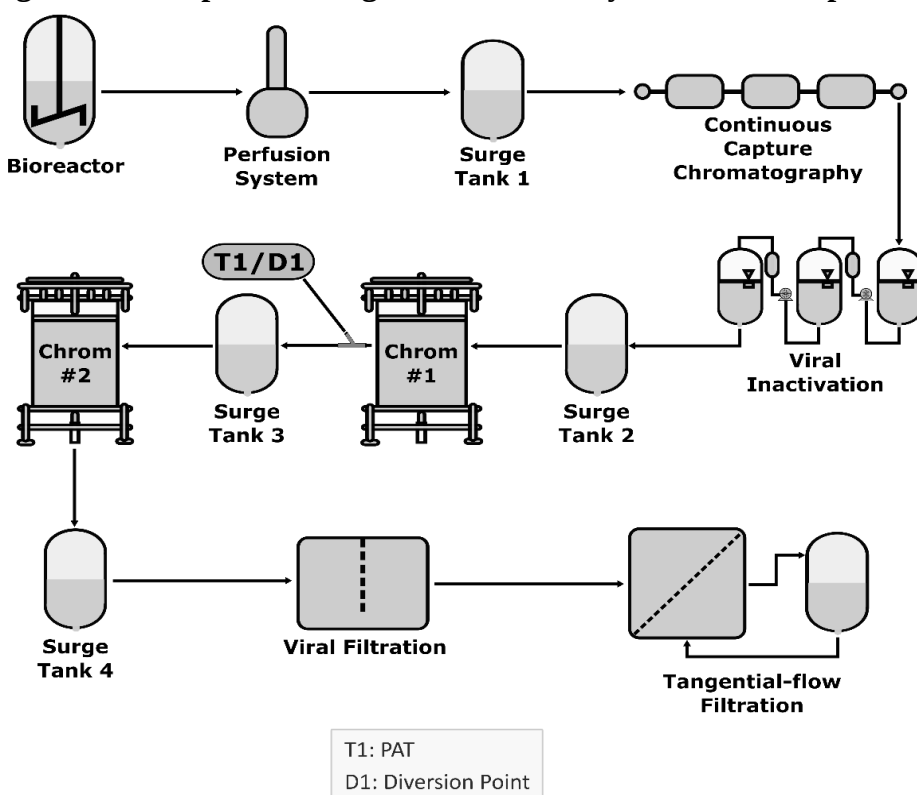
Annex III supplements the main body of this guidance by providing additional regulatory and scientific considerations that are relevant to CM processes for therapeutic protein drug substances and recombinant proteins used as intermediates for subsequent conjugation. It describes aspects that could be applied in fully or partially integrated CM systems. The discussion points and examples presented in the annex are provided for illustrative purposes and are not exhaustive. Alternative approaches can be used.

Figure 3 illustrates a continuous manufacturing process for therapeutic proteins that consists of unit operations characterized by a bioreactor compatible with a perfusion culture system, continuous capture chromatography, virus inactivation, and polishing purification chromatography columns, virus filtration, and buffer exchange and concentration through tangential flow filtration. It is not intended to represent a regulatory flow diagram.

This process integrates a perfusion cell culture with continuous downstream chromatography and other purification steps to continuously capture and purify the target protein. Each individual unit operation is integrated with adjacent unit operations or with a surge line or tank that connects unit operations. Diversion point D1 and PAT (T1) are located after chromatography (Chrom) #1. Using a surge line or tank allows continuous operations to accommodate differences in mass flow rates or process dynamics. Other examples of CM systems may use integrated unit operations for selected steps.

In CM processes, a single thaw of one or multiple vials from the same cell bank may result in either a single harvest or multiple harvests. The number or range of cell bank vials used to produce the specified number of drug substance batches should be defined. The cell bank vials used should be traceable to the output drug substance batches.

Figure 3: Example of a Drug Substance CM System for Therapeutic Proteins



II. CONTROL STRATEGY (2)

A. Adventitious Agent Control (2.1)

In general, all principles used to ensure safety in batch manufacturing are applicable to CM. Safety is demonstrated by a threefold approach based on the principles outlined in the ICH guidance for industry Q5A *Viral Safety Evaluation of Biotechnological Products Delivered From Cell Lines of Human or Animal Origin* (September 1998). Control of adventitious agents (e.g., bacteria, viruses, fungi, mycoplasma) should be based on a risk assessment of all potential sources of contamination (e.g., starting and raw materials, manufacturing operations); the ability of the process to remove and inactivate adventitious agents; and the testing capability to ensure the absence of adventitious agents. Based on this assessment, a strategy should be developed to include the type and frequency of adventitious agent testing undertaken to demonstrate that the process remains free of contamination during cell culture and other downstream steps. An aspect unique to CM is extended cell culture duration and continuous processing of harvested cell culture material to obtain drug substances. This means that design measures should be in place to demonstrate the acceptability of all cell culture material used to generate a given drug substance batch. New technologies for real-time decision making, such as rapid testing for adventitious agents, may be used to mitigate the impact of contamination events by earlier detection and appropriate responses during continuous operation.

B. Equipment Design and System Integration (2.2)

The integrity of single-use equipment during use should be ensured to prevent contamination. Single-use connections (e.g., tube welds, connectors) and components may experience long durations or high change-out frequencies and should be evaluated as potential contamination risks. Filtration steps in CM may be subject to longer filtration periods and potentially increased throughput per unit area or a greater number of filter changes than those under batch manufacturing. Given these factors, a control strategy and a clearly defined scheme should be put in place to allow for filter changes and pre- and post-use integrity testing, as appropriate, without interrupting the process. In the event of a filter failure, a clear strategy for material diversion and refiltration (reprocessing) should be defined.

The CM system should contain appropriate sampling locations based on risk assessment to enable detection of contamination, while avoiding unnecessary contamination risk introduced through the sampling procedure. The sampling locations and frequency may be adjusted based on improved product and process understanding.

Integrated systems may use surge tanks for flow rate adjustments or other purposes between processing steps. When surge tanks are used, the relevant RTD, uniformity and microbial risks to the product in these surge tanks should be evaluated and defined in advance.

C. Process Monitoring and Real-Time Release Testing (2.3)

CM lends itself to various monitoring schemes with different levels of automation. Examples include in-line sensors placed directly in a process vessel or flowing material stream and online analyzers that conduct automatic sampling. Regardless of the approach used, appropriate monitoring at suitable stages of the CM process enables timely data analysis to ensure operations are in a state of control. In certain cases, relevant process parameters can be adjusted to ensure the quality of in-process or output materials. Enhancing in-line/online PAT capabilities and development of automation systems for process monitoring enable a continuous monitoring scheme in support of a release testing strategy that may include RTRT for some quality attributes. For example, drug substance in-line release tests for pH, osmolality, protein concentration and online release tests for purity, charge heterogeneity, aggregation, and low-molecular weight impurities can be performed at specific points in the drug substance manufacturing process shown to be critical for control of the product quality attributes.

An in-process test that is demonstrated to be representative of a release test performed on the output material may be used as a surrogate for a traditional release testing method. Conventional offline testing for product release is necessary for quality attributes for which analytical technologies are not available for online or in-line measurements (e.g., potency). Likewise, conventional tests for monitoring and control (e.g., microbiological analytical methods and other tests that require long processing times) might also be needed.

III. PROCESS VALIDATION (3)

A. Approaches to Process Validation (3.1)

Process validation approaches used for batch processes are also applicable to CM processes with additional considerations for equipment performance over extended run times, integration of automation systems, and material flow throughout a CM system. Therefore, the scope of validation continues to demonstrate the ability of the CM system to consistently manufacture a product with the desired quality.

For therapeutic protein CM, any approach chosen to demonstrate the consistency of process performance and product quality should also consider sources of variability that may have a potential impact on product quality. This may include variability between batches purified from harvest materials collected up to the limit of in vitro cell age from a single cell bank thaw, as well as the potential variability between different batches purified from harvests of multiple cell bank thaws. Variability can be evaluated either as part of process validation or through alternative studies, if justified.

Alternative process validation approaches (e.g., continuous process verification) can be considered when justified. Elements such as risk assessment, applicability of development data, control strategy, and prior knowledge can be considered in determining the suitability of an alternative process validation approach.

B. Run Time Considerations (3.2)

Bioreactors for CM may operate for significantly longer periods of time than bioreactors for batch manufacturing. The approach to establish a limit of in vitro cell age for production cells does not differ, regardless of the mode of bioreactor operation. Previously established limits of in vitro cell age for a bioreactor operating in a batch mode run may not be applicable to a bioreactor operating in a continuous mode under different culture conditions. The limit of in vitro cell age used for production should be based on data derived from production cells expanded under pilot-plant scale or commercial-scale conditions to the proposed in vitro cell age or beyond as outlined in ICH Q5A, the ICH guidance for industry *Q5B Quality of Biotechnological Products* (February 1996) and the ICH guidance for industry *Q5D Quality of Biotechnological/Biological Products* (September 1998).

Run time considerations should include factors such as the control of all adventitious agents (e.g., viruses, bacteria, fungi, mycoplasma) and resin and membrane lifetimes. Viral testing should be conducted as outlined by ICH Q5A, and an appropriate microbial control strategy should be established.

C. Viral Clearance Validation (3.3)

The general recommendations outlined in ICH Q5A for viral safety and clearance remain applicable for CM. When recommendations may not be applicable to a CM system, scientifically justified alternatives can be proposed.

ANNEX IV: INTEGRATED DRUG SUBSTANCE AND DRUG PRODUCT CONTINUOUS MANUFACTURING

I. INTRODUCTION (1)

Annex IV supplements the main body of this guidance by providing additional regulatory and scientific considerations that are relevant for the development and implementation of integrated drug substance and drug product CM processes (referred to as integrated process(es) hereafter).

This annex also provides an example of an integrated process for a small molecule tablet dosage form. The example and approaches described in this annex are not exhaustive. Alternative approaches can be used.

II. INTEGRATED SMALL MOLECULE DRUG SUBSTANCE/DRUG PRODUCT PROCESSES (2)

A. Characteristics of Drug Substance and Drug Product Process Steps (2.1)

Considering the differences between the drug substance and drug product process steps enables appropriate design of an integrated process. For example, process steps for drug substance and drug product manufacturing can have different RTDs, and a prevalence for liquid or solid input material addition can lead to a different frequency of in-process measurements. These differences are expected to influence the selection of equipment, equipment connections, surge lines or tanks, and the locations of in-process measurements and material diversion.

B. Example of an Integrated Process (2.2)

Figure 4, which is not intended to represent a regulatory flow diagram, illustrates a fully continuous integrated drug substance and drug product process. It shows the following elements:

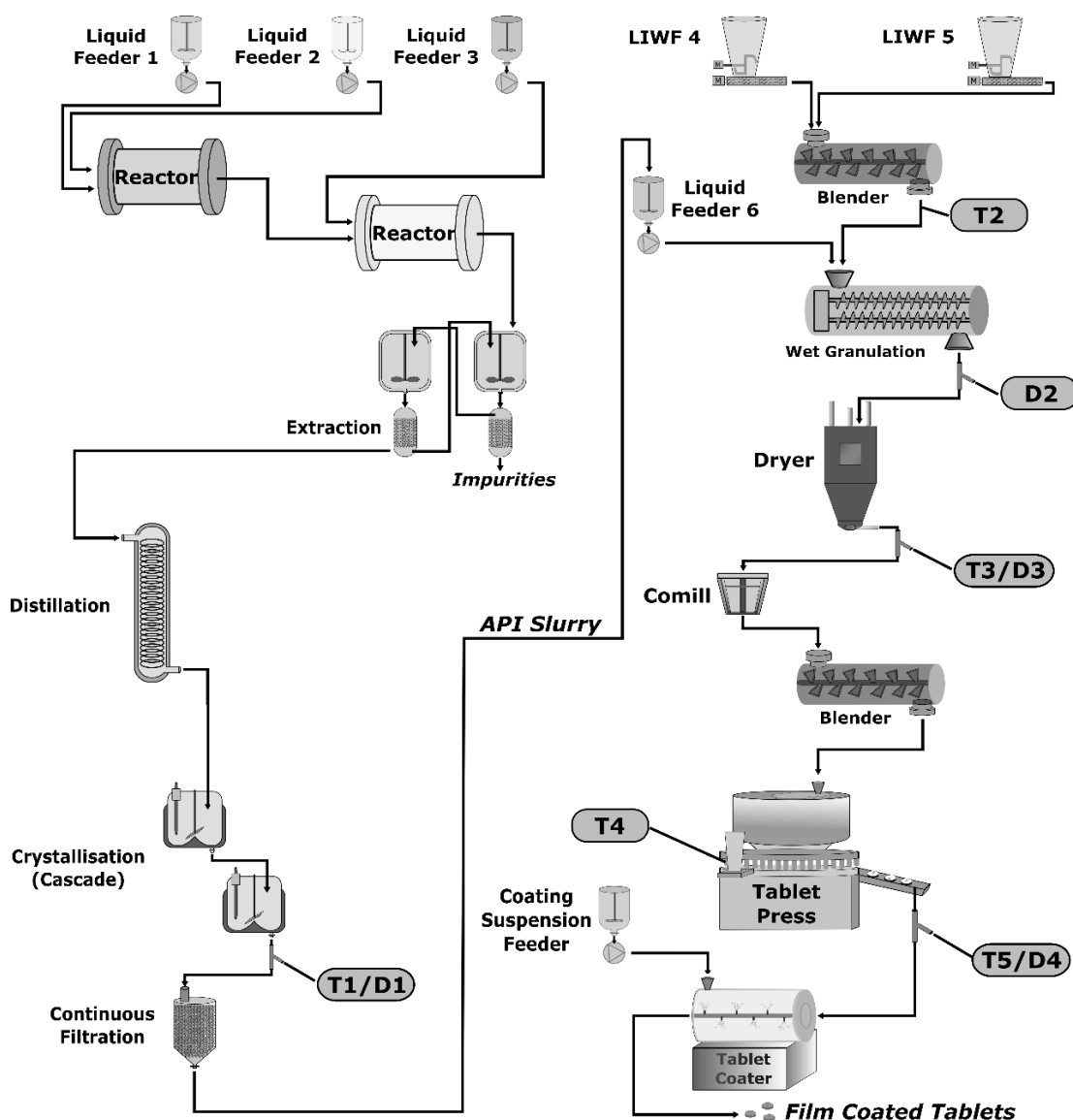
- Material addition points for liquids and solids
- Each process step used for drug substance and drug product manufacturing
- Process design for the interface between the drug substance and drug product
- Sampling locations for all in-line/at-line/offline measurements, including PAT (shown by T1–T5)
- All diversion points (shown by D1–D4)

In this example, chemical reaction using flow reactors, continuous crystallization, and crossflow filtration are used to obtain the drug substance as a highly concentrated crystal slurry. A wet granulation process consisting of blending, granulation, drying, milling, compression and coating unit operations is used to obtain a tablet drug product. The selection of a wet granulation process for the manufacture of the drug product permits the drug substance and drug product processes to

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be integrated through the continuous filtration line. The concentrated crystal slurry functions as both the drug substance source and the granulation fluid. No surge lines or tanks are used. Other process schemes—including, for example, different purification methods, surge tanks, mix of batch and continuous unit operations—could also be used in the design of an integrated process. Details should be provided on how drug substance purity is ensured.

Figure 4: Example of an Integrated Drug Substance and Drug Product CM System



C. Process Design, Monitoring and Control (2.3)

Monitoring points create several process segments, as illustrated in Figure 4 (i.e., from the first drug substance reactor up to location T1, process steps from T1 to T2, etc.). The sampling strategy could be based on RTD characterization of individual steps, process segments, or the entire process. In this example, the RTD of the drug substance process segment provides a suitable time frame to monitor drug substance quality in real time, considering an appropriate sampling frequency, test method, time needed for measurement, and instrument capability. Location T1/D1 is used for sampling drug substance for offline testing or for diversion of drug substance, as necessary. Diversion of material impacts mass flow and may need a compensation strategy in the downstream operations considering the RTD.

Allowable variations (including minor disturbances) identified through DOE or other suitable studies are incorporated into the process control strategy. In this example, the process parameter ranges for material additions and reactors, as well as the magnitude and duration of an allowable disturbance, are based on the variations shown to be within the purification capability of the crystallization step, so there would be no adverse impact on the drug substance purity and impurity profile. An additional risk-based safety margin is included in the established thresholds to ensure all non-conforming material is diverted. Variations outside these thresholds result in material diversion using a suitable method for material traceability (e.g., RTD model).

Ongoing assessment of equipment performance helps predict and prevent potential problems and ensures the ability of a CM process to operate as intended over time. Two such examples are: (1) during continuous filtration, monitoring filter back pressure to evaluate filter saturation (maximum pressure) and prevent filter failure; and (2) during material feeding using LIWFs, monitoring the feeder screw speed in relation to its maximum capacity to inform low feeder fill-level. Monitoring of equipment performance could be used to support how process control will be ensured, especially during long run times.

D. Start-up and Shutdown (2.4)

Individual unit operations of an integrated drug substance and drug product process could achieve desired operating conditions at different times due to differences in the type of transformation (e.g., chemical versus physical transformation) and the residence time in the equipment. When such differences occur, careful planning of start-up and shutdown sequences reduces waste.

E. RTD Characterization for System Dynamics and Material Traceability (2.5)

Refer to the main body of this guidance for RTD characterization. As an integrated process has liquid and solid process streams, different approaches or tracers can be used to characterize various process segments considering the physical state of material in the flow stream (e.g., solution, slurry, solid).

III. SPECIFICATION AND BATCH DATA (3)

A. Drug Substance Specification (3.1)

Even though the drug substance is not isolated in an integrated drug substance and drug product process, a drug substance specification should be defined and justified in accordance with the ICH guidance for industry *Q6A Specifications: Test Procedures and Acceptance Criteria for New Drug Substances and New Drug Products: Chemical Substances* (December 2000) and other relevant ICH guidances. Institution of a drug substance specification defines the quality of the drug substance, as well as facilitates the management of lifecycle activities (e.g., facility changes), investigation of adverse events and product recalls, development of pharmacopeial monographs, and the establishment of a reference standard.

Although a drug substance specification should be instituted, alternatives to drug substance testing on a routine basis can be considered when the integrated process is appropriately controlled. A set of process performance criteria can be defined such that the drug substance could be considered *conforms to specification, if tested* when those process performance criteria are met. To ensure there is a comprehensive monitoring of the quality of the drug substance during the lifecycle of the product, conformance to the drug substance specification should be verified on a periodic and event-driven basis by testing the purified drug substance at an appropriate location using a relevant sampling plan. The frequency of the periodic verification should be defined and justified. Drug substance periodic verification can be based on preselected batches and/or at predetermined time intervals (e.g., every fifth batch or if manufactured infrequently, annually). Event-based verifications could be triggered by a change in supplier, starting material, synthesis conditions, or other factors considering risk. Refer to ICH Q6A for additional details on periodic testing.

Appropriate sampling locations should be incorporated into the process design to enable testing of the drug substance (e.g., location T1 in Figure 4). Any modifications made to the sample to enable the test (e.g., drying of the crystal slurry for testing crystalline form) should be incorporated into the test methodology. Sampling locations should be identified in the drug substance specification.

Although the drug substance is not isolated, the origin and fate of potential impurities (e.g., related substances, residual solvents, catalysts), robustness of impurity clearance, and impurity carryover from the drug substance into the drug product should be provided in the dossier. The control of impurities formation and clearance should be integrated into the overall control strategy.

B. Drug Product Specification (3.2)

In integrated processes, attributes typically associated with the drug substance quality are generally included in the drug product specification unless justified per ICH Q6A. Therefore, the drug product specification in an integrated process should include drug substance-related substances, residual solvents (used in drug substance synthesis), elemental impurities, etc., when appropriate.

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The specified impurities in the drug product specification may differ from the specified impurities in the drug substance specification.

Sampling location should be appropriately identified in the drug product specification table because some testing (e.g., testing for drug substance periodic verification as described above) may need to be performed following the drug substance purification step (before drug product formation).

C. Example of a Drug Substance and Drug Product Specification (3.3)

An example of a drug substance and drug product testing approach for an integrated process based on Figure 4 is shown in Table 4. The test attributes listed are considered relevant for this example. The specific details of each integrated process should be considered in the selection of the appropriate test attributes and testing plan.

Table 4: Example of a Testing Approach for an Integrated CM

Test Attribute ¹	Drug Substance Specification for Periodic Testing		Drug Product Specification for Routine Testing of Every Batch	
	Test	Sampling Location	Test	Sampling Location
Description	N/A	N/A	✓	Coated tablets
Identity	✓	Use drug product test result	✓	PAT at tablet feed frame (T4)
Crystalline Form ²	✓	Sampling Location T1	N/A	Not tested when justified
Chirality ³	✓	Sampling Location T1	N/A	Not tested when justified
Particle Size	✓	Sampling Location T1	N/A	Not tested
Purity	✓	Sampling Location T1	N/A	Not tested
Assay	N/A	N/A	✓	Core tablets, Sampling Location combination of T4 (blend uniformity) and T5 (tablet weight)
Impurities ⁴	<i>Impurity specification for drug substances and drug products may differ</i>			
<i>Related Substance</i>	✓	Sampling Location T1 (at-line high performance liquid chromatography (HPLC)) ⁴	✓	Sampling Location T1 (at-line HPLC) ⁴ or Coated Tablets (offline HPLC testing), as appropriate
<i>Residual Solvents</i>	✓		✓	
<i>Elemental Impurities</i>	✓		✓	
<i>Mutagenic Impurities</i>	✓		✓	
Dissolution	N/A	N/A	✓	Coated Tablets
Uniformity of dosage units	N/A	N/A	✓	Uncoated Tablets
Water content	N/A	N/A	✓	Coated Tablets
Microbial limits	N/A	N/A	✓	Coated Tablets

¹ Include tests that are critical to ensure the identity, strength, quality and purity of the drug substance and bioavailability of the drug product as per ICH Q6A.

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² In this example, crystalline form is considered a critical quality attribute for the drug substance and hence tested periodically. Crystalline form is not tested in the drug product because lack of form change during drug product processing has been demonstrated.

³ In this example, chirality is considered a critical quality attribute for the drug substance.

⁴ Tests that are common to both drug substance and drug product specification can be tested only at one location; the same test result can be used for the drug substance and drug product.

D. Batch Data (3.4)

Although the drug substance is not isolated, small, planned diversions during process development should be used to obtain batch data that is representative of commercial drug substance.

IV. STABILITY (4)

A. Drug Substance Stability (4.1)

Drug substance stability data to define a retest period is not applicable because the drug substance is not isolated and stored in an integrated process. Drug substance stability data may be appropriate for other aspects, such as to support the storage of in-house reference standards and to gain an understanding of product stability profiles. Institution of a hold time enables temporary storage of drug substance during an interruption in production.

B. Drug Product Stability (4.2)

The ICH stability guidances and section IV.E (4.5) are applicable to drug product made by an integrated process.

V. LOCATION OF DRUG SUBSTANCE AND DRUG PRODUCT INFORMATION IN THE CTD (5)

Drug substance and drug product information could be provided in the respective CTD sections 3.2.S and 3.2.P of the dossier. A description of the process step that integrates the drug substance and drug product could be based on its relevancy to the respective section. For example, in the process example provided in this annex, the continuous filtration process could be described in CTD section 3.2.S because it is related to concentration of the drug substance. The integrated flow diagram can be provided in CTD section 3.2.P and referenced in section 3.2.S.

ANNEX V: PERSPECTIVES ON MANAGING DISTURBANCES

I. INTRODUCTION (1)

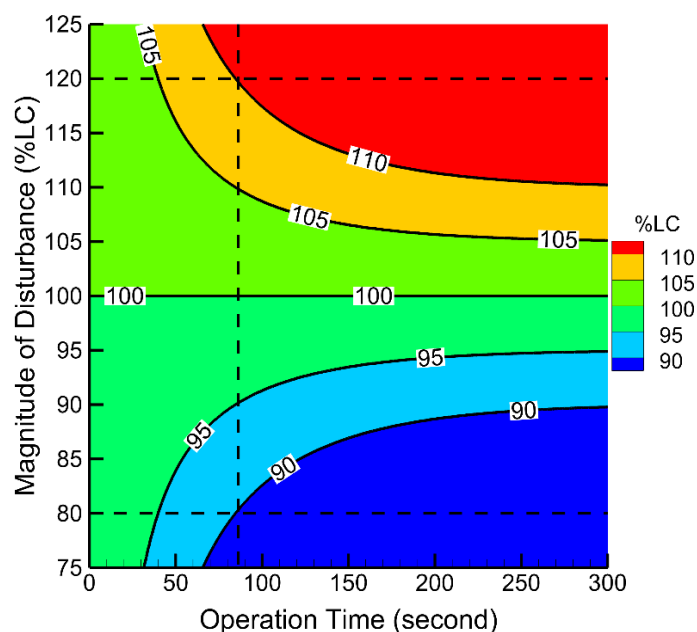
This annex describes examples of approaches for managing transient disturbances (hereafter referred to as disturbances in this annex) that may occur during CM. The discussion points presented here are not exhaustive. Alternative approaches can be used.

II. BACKGROUND (2)

Disturbances may result in a variation in material quality. The impact of some variations on material quality in an earlier process step may be resolved by downstream process steps. The extent of variations and the ability to resolve them in subsequent steps are impacted by the amplitude, duration, and frequency of the disturbance. Identification of tolerable ranges for these parameters and establishing appropriate acceptance criteria will enable the development of an effective strategy for managing disturbances.

Manufacturers can use various methodologies (e.g., DOE, RTD studies, a combination of both) to understand the impact of disturbances. Funnel plot predictions based on an RTD model can be a useful tool to understand the qualitative and quantitative impact of the amplitude and duration of a disturbance on material quality. Figure 5 shows a funnel plot for drug substance feeding in a drug product CM process (similar to the example in Annex II). Funnel plots are specific to the formulation, process conditions, and system configuration used in RTD model development. Information from the funnel plot helps to inform the selection of appropriate acceptance criteria for disturbances. For example, the dotted lines in the following funnel plot show that a disturbance of ± 20 percent lasting less than 90 seconds would not cause the drug concentration in the blend to exceed the 90 to 110 percent label claim (LC).

Figure 5: Example of a Funnel Plot for the Feeding of a Drug Substance



III. MANAGEMENT OF DISTURBANCES (3)

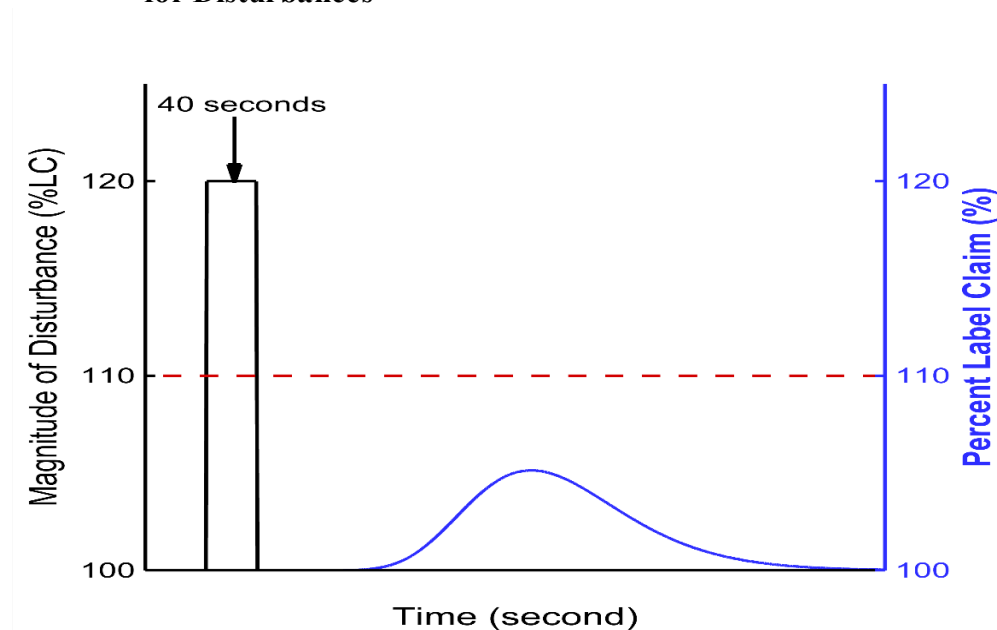
Manufacturers can develop various approaches to manage disturbances considering the specific details of the CM system and the risk to material quality from disturbances. Three examples considering different risks of a disturbance are provided below:

- Example 1: The amplitude and duration of the disturbance meet predefined acceptance criteria for the disturbance, and the occurrence of such disturbances is infrequent.
- Example 2: The amplitude or duration of the disturbance exceeds the predefined acceptance criteria for the disturbance, and the occurrence of such disturbances is infrequent.
- Example 3: The amplitude and duration of each disturbance meet predefined acceptance criteria for the disturbance, but multiple, frequent disturbances are observed.

These examples focus on the impact of disturbance from a loss-in-weight feeder (LIWF) on the drug concentration in the blend for a CM process similar to that described in Annex II, given that all other parameters being monitored meet the predefined acceptance criteria. These examples use the information in the funnel plot (Figure 5) and, for the purpose of discussion, assume that the acceptance criteria for the magnitude and duration of an LIWF disturbance is ± 20 percent lasting for 80 seconds. These examples help illustrate the important considerations in management of disturbances under selected scenarios, which may also be applicable to drug substances and other CM processes.

A. Disturbance Example 1 (3.1)

Figure 6: Example of an Infrequent Disturbance That Is Within the Acceptance Criteria for Disturbances



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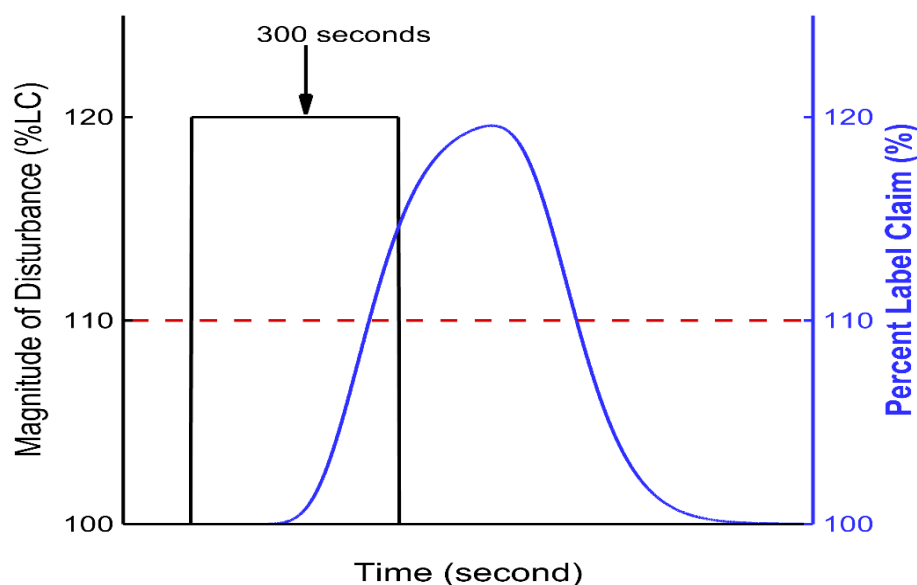
Description: Figure 6 illustrates a drug substance LIWF with an infrequent transient +20 percent flow spike lasting 40 seconds, which is within the predefined acceptance criteria for disturbances. This disturbance causes an increase in the amount of the drug substance fed into the blender, before returning to normal operating condition. The funnel plot (Figure 5) shows that following this disturbance, the drug substance concentration in the blend remains within the 90 to 110 percent acceptance criteria, due to back mixing. An additional quality check, such as measurement of the drug substance concentration at a suitable location (e.g., NIR measurements at the tablet press feed frame) could be considered, to confirm the blend is within 90 to 110 percent.

Impact: Although this disturbance represents an excursion from normal operation, the quality of the output material is not affected as the magnitude/amplitude of the disturbance and product quality meet their predefined acceptance criteria.

Action: No material is diverted. Collection of the output material continues, and the process continues to operate. No investigation is needed, because such a disturbance has been evaluated during development and impact on material quality is understood.

B. Disturbance Example 2 (3.2)

Figure 7: Example of an Unexpected Disturbance That Is Outside the Acceptance Criteria for Disturbances



Description: Figure 7 illustrates a drug substance LIWF with an unexpected transient +20 percent flow spike lasting 300 seconds. The disturbance is outside the predefined acceptance criteria for disturbances. This disturbance causes an increase in the amount of the drug substance fed into the blender before returning to normal operating condition. The funnel plot (Figure 5) shows that following this disturbance, the drug substance concentration in the blend exceeds the

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90 to 110 percent acceptance criterion. An additional quality check, such as measurement of the drug substance concentration at a suitable location (e.g., NIR measurements at the tablet press feed frame), confirms the blend exceeds 110 percent.

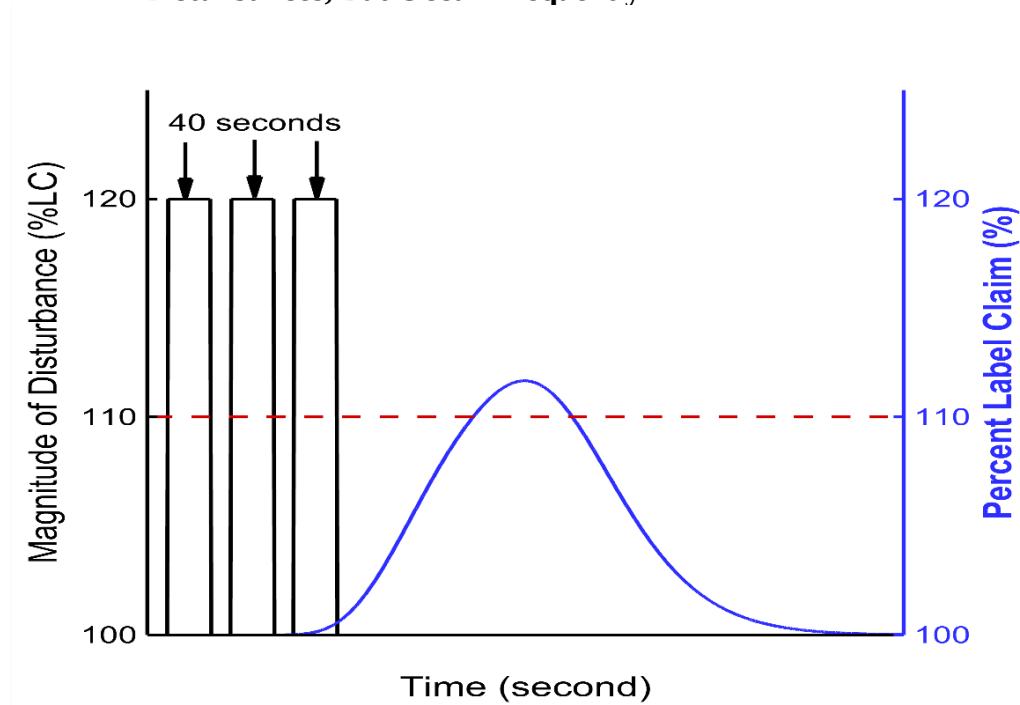
Impact: The quality of the output material is adversely impacted because the disturbance duration exceeds the predefined acceptance criteria.

Action: The process continues to operate while the nonconforming material is diverted according to a preestablished procedure, and the time to start and end diversion is controlled by the automation system. The system returns to normal material collection mode when the nonconforming material is completely diverted. If needed, a concurrent investigation is initiated to determine root cause.

Diverted Amount: The amount of material diverted depends on the control strategy used (including specific triggers for material diversion) and on the process dynamics from the point of disturbance detection and the point at which material diversion ends. Inclusion of confidence intervals in the RTD provides a safety margin to ensure all nonconforming material is diverted from the batch. Additional factors, such as the sampling strategy and the ability to trace and remove materials, are considered in establishing the criteria for material diversion.

C. Disturbance Example 3 (3.3)

Figure 8: Example of Disturbances That Are Within the Acceptance Criteria for Disturbances, But Occur Frequently



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Description: Figure 8 illustrates a drug substance LIWF with multiple frequent transient +20 percent flow spikes, each lasting 40 seconds, resulting in variability in the amount of material fed into the blender.

Impact: Although each disturbance meets the predefined acceptance criteria for disturbances, the disturbances occur with a high frequency over a short time period. In this example, the system cannot dampen these multiple disturbances sufficiently, thus resulting in nonconforming materials.

Action: The impact of these disturbances on system performance and output material quality is monitored closely (e.g., NIR method, other elements of the control strategy). Process operation and product collection continue until one or more elements of the control strategy do not meet the predefined acceptance criteria. When a criterion is no longer met, the material is diverted according to a preestablished procedure. If high-frequency disturbances persist, process operation may be paused. An investigation is conducted to understand the root cause for these frequent disturbances. Such investigations enable preventative actions to be taken to avoid equipment failure and adverse impact on critical quality attributes, to ensure process performance (e.g., robustness), etc. Assessment of process capability or other evaluations may also be warranted. Setting acceptance criteria for the frequency of disturbances could also be considered to aid the management of disturbances.

Diverted Amount: The amount diverted is the same as described in section III.B (3.2) Annex IV. The disposition of the diverted material and the entire batch is assessed upon completion of the investigation.